IGP or Integrated Graphics Processor. The IGP will have an inbuilt graphics accelerator, the likes of a GeForce 2 MX, to power your graphics applications and 3D games, and aspires to be the most powerful integrated graphics chipset ever put on a system, with a filter rate of 350 mega pixels per second. The Geforce 2 MX also includes standards and features such as Transform and Lighting (TnL), which constitutes live rendering of paths of light in games, 3D applications, etc.

While there won't be any integrated RAM for the graphics card, it will make use of your system memory for applications. And if you're in search of the best 3D accelerator card around, you can always plug-in an external 3D accelerator card into the AGP slot on the motherboard, unlike boards such as the i810. The integrated Geforce 2 MX gives you the advantage of 6x AGP bandwidth access to the motherboard.

Another technology integrated in the IGP, called the TwinBank Memory Architecture, gives high bandwidth access between the graphics processor and memory, equivalent to a DDR (Double Data Rate) memory based 3D accelerator.

MCP: The South Bridge on the nForce-based motherboards is also called the Media and Communications Processor (MCP), which earlier used to handle the IDE controllers and various interfaces such as the PCI Slot, etc. The nForce chipset will come with a dual ATA-100 controller for ATA 100 hard drives. Coupled with this is a built-in 10/100 MBps LAN card with an internal 56 Kbps modem. There are also going to be six USB ports instead of the normal two to four ports on standard motherboards.

Streamthru is another technology that will provide high bandwidth between various interfaces such as disk drives, LAN, etc, thereby giving you smooth audio and video coupled with network speed enhancements.

APU: Generally, you would have to buy a surround soundcard, which would give you 4.1 or 5.1 surround sound. One of the main additions to the integrated chipsets is the Audio Processor Unit or APU. While most integrated chipsets give you simple stereo sound, the APU in the nForce chipset comes with 5.1 surround sound, thus converting your computer into a personal home theatre system—provided, of course, that you get yourself a good set of surround speakers. The APU is totally DirectX 8 compliant and gives you 64 3D streams or 256 stereo streams.

The technologies involved

Dynamic Adaptive Speculative Pre-Processor (DASP): This is a smart piece of technology whose basic task is to ascertain future patterns that may be encountered by the processor. This takes the load off your main CPU which otherwise handles this task. The technology allows you to make use of the bandwidth, which may be static at any particular moment, to ascertain future information and cache the same for later use, thereby giving optimal performance. Similar technology has been implemented in the Intel Pentium 4 processor for improved performance. However, among motherboards, this technology is available only on the nForce.

TwinBank Memory Architecture: This is a new technology used in the nForce's IGP for providing very high bandwidth. There are two types of TwinBank Memory Architectures: the 128-bit and 64-bit 133 MHz DDR interface. Intel's Pentium 4 uses a similar method using a dual RDRAM (Rambus Dynamic RAM) channel while the nForce uses DDR RAM. Though cheaper than the RDRAM, it is priced quite high even now.

AMD Hypertransport: nVidia has brought into action the AMD Hypertransport, earlier known as Lightning Data Transport. In the nForce chipset, the AMD Hypertransport technology comes into use between the IGP and MCP. The width of the ports used for sending and receiving data from one chipset to another on the nForce-based motherboard is anything between 2 to 32 bits, operating at a frequency of 200 MHz. This runs at a double rate, thus making the transfer rates similar to a 400 MHz bus, implying plenty of bandwidth between the chipset itself.

Getting them going

So how do all these high-end components, integrated into your motherboard, work together to

Specification	Regular system		With nForce-basse	
	Configuration	Price	Configuration	Price
CPU	Intel 1GHz	$170	AMD Athlon 1GHz	$90
Motherboard	815 Motherboard	$90	NForce-based motherboard	$200
Graphics	GeForce 2 MX	$50	Integrated	-
Sound	SbLive Value	$40	Integrated	-
Modem	56kbps modem	$10	Integrated	-
Ethernet	10/100MBPS NIC	$10	Integrated	-
RAM	128MB SDRAM	$10	128MB DDR RAM	$20
Total		Approx $380		Approx $310

Versions of the nForce

There will be quite a few variants of the nForce chipset available, with all kinds of add-on and reducible components as per individual needs. The IGP could have a 64-bit memory interface or a 128-bit memory interface. As for the MCP, the Audio processor can be Dolby Digital 5.1, or without the Dolby Digital sound for those who don't need it. The prices for these additions and reductions will also change accordingly. You can also decide on an integrated graphics processor or the plain motherboard itself with the rest of the features. So if you have a higher graphics card than the GeForce 2 MX, such as a GeForce 2 GTS or a GeForce 3, you could decide on an nForce-based motherboard without the graphics processor, thus cutting costs. Similarly, if you have an SB Live soundcard and need to upgrade to a higher configuration, but without the Dolby Digital 5.1 sound from the integrated APU, then you can choose the chipset with the integrated APU.